

Evaluation and Observability Strategy for Real Estate AI Platform

Objective

Building an AI platform does not end with ingestion and retrieval.

A production-grade AI system must continuously answer the following questions:

1. Is the system returning the correct information?
2. Is the retrieved context relevant?
3. Is the LLM hallucinating?
4. Is the system meeting latency and cost objectives?
5. Can we quickly identify failures and bottlenecks?
6. Are users satisfied with the generated answers?

To achieve this, I would design both:

- Evaluation Framework
- Observability Framework

Evaluation measures quality.

Observability explains why quality changed and where failures occurred.

Evaluation Framework

I divide evaluation into four layers:

1. Data Quality Evaluation
2. Retrieval Quality Evaluation
3. Generation Quality Evaluation
4. User Satisfaction Evaluation

Layer 1: Data Quality Evaluation

Garbage In → Garbage Out

If ingestion quality is poor, no retrieval or LLM can compensate.

Freshness Evaluation

Goal:

Ensure latest document versions are searchable.

Examples:

- Latest Sale Deed indexed
- Latest Property Update indexed
- Updated Price reflected in search

Metrics:

- Indexing Delay
- Stale Document Rate
- Version Mismatch Rate

Example:

Document Version = V10

Retrieved Version = V8

This indicates stale indexing.

Parsing Accuracy

Measure quality of extraction.

Metrics:

- OCR Accuracy
- PDF Parsing Accuracy
- Table Extraction Accuracy
- Image Understanding Accuracy

Examples:

- Missing legal clause
 - Missing property dimensions
 - Missing ownership information
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Chunk Quality

Evaluate:

- Chunk Size Distribution
- Context Preservation
- Semantic Integrity

Questions:

- Are chunks too small?
 - Are chunks too large?
 - Did chunk boundaries break meaning?
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Metadata Accuracy

Metadata directly impacts retrieval quality.

Examples:

- Location
- Property Type
- Price
- Bedrooms
- Tenant ID
- Document Version

Metrics:

- Metadata Extraction Accuracy
 - Metadata Completeness
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Layer 2: Retrieval Quality Evaluation

The purpose is to determine whether the correct information is being retrieved.

Recall@K

Question:

Was the relevant document retrieved?

Example:

Top 10 Retrieved Documents

Relevant document exists?

Metric:

Recall@10 Recall@20

Higher is better.

Precision@K

Question:

How much irrelevant content was retrieved?

Example:

100 retrieved chunks

Only 5 are relevant

Precision is poor.

Metric:

Precision@10 Precision@20

Mean Reciprocal Rank (MRR)

Measures ranking effectiveness.

Question:

How early does the relevant document appear?

Example:

Rank 1 = Excellent

Rank 20 = Poor

NDCG

Normalized Discounted Cumulative Gain

Measures ranking quality when multiple relevant documents exist.

Useful for:

- Property search
 - Property comparison
 - Legal document retrieval
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Context Utilization

Question:

Did the LLM actually use retrieved context?

Example:

Retrieved:

50 chunks

Used:

3 chunks

This indicates inefficient retrieval.

Metrics:

- Retrieved Context Size
 - Context Utilization Rate
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Parent-Child Expansion Evaluation

For hierarchical chunking:

Evaluate:

- Expansion Success Rate
- Expansion Relevance
- Additional Token Cost

Question:

Did parent expansion improve answer quality?

Or only increase context size?

Layer 3: Generation Quality Evaluation

Evaluates final LLM output.

Faithfulness

Most important metric.

Question:

Is the answer grounded in retrieved documents?

Or generated from model memory?

Metrics:

- Faithfulness Score
 - Groundedness Score
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Hallucination Rate

Question:

Does the answer contain unsupported claims?

Metric:

Hallucination Percentage

Goal:

As close to zero as possible.

Answer Relevance

Question:

Did the answer address the user's query?

Metric:

Answer Relevance Score

Completeness

Question:

Did the answer include all relevant information?

Example:

User asks:

"What transfer restrictions apply?"

Answer includes only one restriction while three exist.

Metric:

Completeness Score

Coherence

Question:

Does the answer follow logical flow?

Example:

Step 1

Step 4

Step 2

Poor coherence.

Metric:

Coherence Score

Citation Accuracy

Question:

Can every claim be traced to source documents?

Metrics:

- Citation Coverage
- Citation Accuracy

Enterprise systems should provide traceability.

Layer 4: User Satisfaction Evaluation

Ultimately, users determine success.

Explicit Feedback

Examples:

- Thumbs Up
- Thumbs Down
- Rating Score

Metrics:

- Positive Feedback Rate
 - Negative Feedback Rate
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Implicit Feedback

Behavioral signals often reveal more than ratings.

Examples:

- User reformulates query
- User searches again immediately
- User abandons answer
- User requests human agent

Metrics:

- Query Reformulation Rate
 - Follow-Up Query Rate
 - Session Abandonment Rate
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Task Completion Metrics

Business-focused evaluation.

Examples:

Property Search:

- Property Clicked
- Agent Contacted
- Visit Scheduled

Legal Search:

- Question Resolved
- Escalated to Human Expert

Metrics:

- Task Completion Rate
 - AI Resolution Rate
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Cost and Latency Evaluation

Production systems must balance quality with efficiency.

Latency Breakdown

Measure each component independently.

Components:

- Query Planner
- Embedding Generation
- Vector Search
- Hybrid Search
- RRF
- Reranker
- Context Builder
- LLM
- Guardrails

Metrics:

- P50 Latency
 - P95 Latency
 - P99 Latency
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Cost Metrics

Track:

- Cost Per Query
- Cost Per Tenant
- Cost Per Model
- Cost Per Feature

Examples:

Property Search = \$0.002

Legal QA = \$0.05

Observability Framework

Evaluation tells us whether the system is good.

Observability tells us why it is good or bad.

I divide observability into five pillars:

1. Data Observability
 2. Retrieval Observability
 3. LLM Observability
 4. System Observability
 5. Business Observability
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1. Data Observability

Monitor ingestion pipeline health.

Metrics:

- Documents Ingested Per Hour
- Parsing Success Rate
- OCR Success Rate
- Embedding Generation Success Rate
- Chunk Count Distribution
- Metadata Extraction Accuracy
- Indexing Delay

Alerts:

- Parsing failure spikes
 - OCR accuracy drops
 - Stale document indexing
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2. Retrieval Observability

Monitor search quality.

Metrics:

- Recall@K
- Precision@K
- MRR
- NDCG
- Retrieval Latency
- Metadata Filter Latency
- Context Expansion Latency

Questions:

- Are relevant documents retrieved?
 - Are filters slowing searches?
 - Is retrieval quality degrading?
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3. LLM Observability

Monitor model behavior.

Metrics:

- Prompt Tokens
- Completion Tokens
- Total Tokens
- Cost Per Request
- Hallucination Rate
- Faithfulness Score
- Citation Coverage
- Model Latency

Model Routing Metrics:

- Small Model Usage
- Medium Model Usage
- Large Model Usage

Example:

80% traffic routed to premium model

Potential cost issue.

4. System Observability

Traditional distributed system monitoring.

Metrics:

API Layer:

- Requests Per Second
- Success Rate

- Error Rate
- Timeout Rate

Infrastructure:

- CPU Usage
- Memory Usage
- GPU Utilization
- Disk Usage
- Network Throughput

Vector Database:

- Query Latency
- HNSW Graph Size
- Memory Consumption
- Index Growth

SQL Database:

- Slow Queries
- Connection Pool Usage
- Lock Contention
- Index Hit Rate

5. Business Observability

Monitor business outcomes.

Metrics:

User Activity:

- Daily Active Users
- Monthly Active Users
- Queries Per User

Trust Metrics:

- Citation Click Rate
- Answer Acceptance Rate
- Feedback Score

Business Metrics:

- Property Click Rate

- Agent Contact Rate
 - Property Visit Rate
 - Legal Resolution Rate
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AI-Specific Alerting Strategy

Freshness Alert

Trigger:

Document uploaded but not indexed within SLA.

Example:

Uploaded > 10 minutes ago

Not searchable

Alert.

Retrieval Quality Alert

Trigger:

Recall@10 drops significantly.

Example:

95% → 80%

Alert.

Hallucination Alert

Trigger:

Faithfulness Score decreases below threshold.

Example:

98% → 85%

Alert.

Cost Alert

Trigger:

Daily spend exceeds expected threshold.

Example:

Expected = \$500/day

Actual = \$1500/day

Alert.

Latency Alert

Trigger:

P95 latency exceeds SLA.

Example:

Target = 2 seconds

Actual = 5 seconds

Alert.

End-to-End Traceability

Every user request should be traceable.

For each query, capture:

- User Query
- Query Plan
- Retrieved Documents

- Retrieval Scores
- Reranking Scores
- Context Used
- Prompt Version
- Model Version
- Generated Answer
- Latency
- Cost

This enables rapid debugging and root-cause analysis.

Final Recommendation

A production-grade Real Estate AI platform should continuously evaluate and observe every stage of the pipeline.

Evaluation focuses on:

- Data Quality
- Retrieval Quality
- Generation Quality
- User Satisfaction

Observability focuses on:

- Data Pipeline Health
- Retrieval Behavior
- LLM Performance
- System Reliability
- Business Outcomes

The ultimate goal is to make every answer explainable, measurable, traceable, and continuously improvable while maintaining latency, cost, and accuracy targets.